

$2\hat{I}$	$\hat{U}^\dagger(t_1, t_0)$	0	0
$\hat{U}(t_1, t_0)$	$2\hat{I}$	$\hat{U}^\dagger(t_2, t_1)$	0
0	$\hat{U}(t_2, t_1)$	$2\hat{I}$	$\hat{U}^\dagger(t_3, t_2)$
0	0	$\hat{U}(t_3, t_2)$	$\hat{I}$